Dual-Stage Prior-Driven Diffusion Model for Remote Sensing Spectral Super-Resolution

Zengyi Li, Junyu Gao, *Member, IEEE*, and Yuan Yuan, *Senior Member, IEEE*

***Abstract*—Spectral super-resolution (SSR) is a key technology for generating high-spatial-resolution hyperspectral images (HSIs). However, deep learning approaches for SSR, especially generative models like diffusion, often rely heavily on large training datasets. Furthermore, their stochastic generation process can compromise the precise spectral fidelity required in remote sensing. To address this limitation, we propose a dual-stage prior-driven diffusion model (DPDM) for SSR tasks. The DPDM comprises two modules: the prior-driven diffusion module (PDM) and the spectral refinement module (SRM). The PDM replaces the conventional pure noise input with a structured prior, which we term the prior-informed noise (PIN). This PIN is deterministically generated by projecting the input multispectral image (MSI) onto a spectral basis, which is extracted from the spectral response function (SRF). By initializing the reverse process with this information-rich starting point, our model significantly reduces its dependence on large training datasets and inherently enforces spectral consistency. The SRM is subsequently introduced to specifically target and correct residual artifacts and coarse features from the initial stage. Employing a hierarchical multiscale architecture and a single-sample optimization framework, the SRM meticulously restores fine-grained details while suppressing noise in the PDM's output. By integrating these two modules, the DPDM progressively enhances both spatial and spectral fidelity. Extensive experimental results demonstrate that the DPDM achieves competitive performance in SSR tasks.**

***Index Terms*—Data generation, diffusion model, dual-stage network, hyperspectral image, spectral super-resolution (SSR).**

I. Introduction

With the advancement of satellite imaging sensors, satellites can capture diverse data modalities, including but not limited to multispectral images (MSIs), hyperspectral images (HSIs), and synthetic aperture radar (SAR). The distinct properties of these data garnered substantial research interest. Specifically, HSIs are characterized by their broad spectral coverage, spanning visible, near-infrared, and long-wave infrared wavelengths, making them suitable for various applications. For instance, HSIs are used in tasks like spectral unmixing [1], [2], change detection [3], [4], data fusion [5], [6], [7], semantic segmentation [8], [9], and object detection [10], [11].

Nevertheless, due to the inherent limitations of their imaging process, the acquisition of HSIs suffers from high costs and low spatial resolution [12]. These drawbacks hinder the widespread application of HSIs. In contrast, MSIs have higher spatial resolution but contain fewer spectral bands, resulting in lower spectral resolution. To overcome these problems, many researchers utilized image fusion methods, which improve the spatial resolution of an HSI by using a paired MSI [13], [14], [15], [16], [17]. Another promising direction, which avoids the need for paired images, is spectral super-resolution (SSR). SSR technology directly generates HSIs from MSIs, thus showing greater feasibility in scenarios where paired data is unavailable. Given its broader applicability, our work focuses on the SSR paradigm.

SSR is a highly ill-posed, underdetermined problem, and numerous solutions exist to address it. Early approaches were predominantly based on spectral sparse priors [18]. These methods involve learning a hyperspectral dictionary and subsequently formulating the MSI-to-HSI conversion as a sparse coding problem [19], [20], [21]. Dimensionality reduction techniques are also widely adopted to model nonlinear intermodal relationships [22]. Nevertheless, sparse coding-based methods are often scenario-specific and suffer from performance degradation in complex environments due to poor generalization. Although these methods possess a solid theoretical foundation, their scene-specific design and reliance on block-wise image processing often limit their generalization capability and ability to extract global features. In contrast, deep learning-based methods typically exhibit superior performance in these respects. Some advanced visual methods [23], [24], [25] emerged in recent years have also been applied to SSR tasks. These approaches are broadly categorized into two groups: the first category comprises discriminative models, with specific implementation techniques based on either convolutional neural networks (CNNs) [26], [27], [28] or Transformers [29]. The goal of these methods is to minimize the error between the generated samples and the target data. While they feature a clear and optimizable objective function, they possess certain limitations in capturing detailed spectral information. The second category consists of generative models, for which the latest paradigms include generative adversarial networks (GANs) [25], [30] and diffusion models. The goal of these methods is to generate samples consistent

Received 20 April 2025; revised 20 July 2025; accepted 10 August 2025. Date of publication 21 August 2025; date of current version 23 September 2025. This work was supported in part by the National Key Research and Development Project under Grant 2024YFC3015504 and in part by the Natural Science Foundation of China under Grant 62306241 and Grant 62576284. *(Corresponding author: Yuan Yuan.)*

The authors are with the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China (e-mail: lizengyi@mail.nwpu.edu.cn; gjy3035@gmail.com; y.yuan1.ieee@gmail.com).

Digital Object Identifier 10.1109/TGRS.2025.3601119

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:23:09 UTC from IEEE Xplore. Restrictions apply.

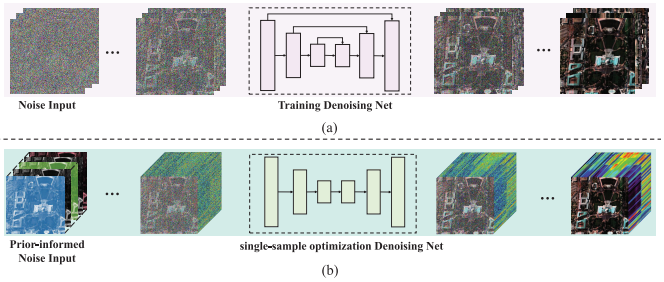


Fig. 1. Comparison between the classical diffusion model and our proposed PDM. (a) Classical diffusion process. In the classical diffusion model, the noise prediction network requires extensive data training. (b) Proposed diffusion process. The PDM achieves SSR from a single sample by predicting noise through a carefully designed single-sample optimization network. Additionally, PIN is introduced to enhance the quality of the generated output, resulting in improved performance.

with the target data distribution. Among these, GANs exhibit good generation quality, but their training process can be stochastic, leading to unstable results.

Diffusion models have achieved superior performance in various vision tasks such as image generation [31], text-to-image generation [32], and super-resolution [33]. These models iteratively remove noise through a reverse diffusion process, enabling precise control over output refinement at each step. However, applying diffusion models to HSI generation introduces critical challenges. The scarcity of HSI training data compared to MSIs exacerbates dataset-induced biases during generation. Furthermore, the diffusion process introduces noise that compromises the reconstruction of intricate spectral details. Consequently, the core challenge lies in harnessing the generalization power of diffusion models to counteract data biases while minimizing spectral distortions caused by noise.

Motivated by these challenges, we propose a dual-stage prior-driven diffusion model (DPDM) for SSR tasks. As illustrated in Fig. 1, the DPDM diverges from conventional diffusion frameworks by introducing two innovations. In the first stage, the DPDM integrates a single-sample optimized noise prediction network with prior-informed noise (PIN), thereby eliminating the reliance on large-scale training datasets. The second stage also utilizes a single-sample optimized, multiscale architecture to compensate for the spectral detail lost in the first stage, iteratively suppressing noise-induced spectral distortion.

The key contributions of this work are summarized as follows.

1) We propose a DPDM, consisting of a PDM and a spectral refinement module (SRM). Owing to its single-sample optimization structure, which requires only a single MSI for processing, the DPDM is particularly well-suited for SSR tasks and demonstrates robust generalization capability.
2) We design a prior-driven diffusion module (PDM) that initializes the reverse process with information-rich PIN instead of pure noise, fundamentally guiding the generation toward a more accurate reconstruction. Additionally, the integration of a redesigned noise prediction network further improves the reconstruction performance.
3) We construct an SRM to reduce the noise introduced by the diffusion model. Its multilayer structure effectively mitigates the noise's impact on the multiscale spatial–spectral structure, ensuring more accurate spectral representation.

II. Related Work

In this section, we introduce the SSR background and technologies involved in the DPDM. This section is divided into two parts: SSR and diffusion-based methods.

A. Spectral Super-Resolution

SSR is an image reconstruction approach designed to enhance the spectral resolution of MSIs [19], [34]. However, differences in imaging modalities between MSIs and HSIs result in substantial structural disparities, which have long posed significant challenges for SSR.

Initial approaches to the SSR problem are rooted in signal-processing theory, with sparse representation over a spectral dictionary emerging as a dominant paradigm. Early works, such as that by Aharon et al. [35], constructed the basic diction using simple transformations, for instance, the pseudo-inverse of the camera's spectral response function (SRF). To better capture the complex characteristics of real-world spectral data, subsequent methods shifted focus to learning the dictionary directly from data. Arad and Ben-Shahar [19] proposed creating a comprehensive dictionary from a large corpus of HSIs, from which a corresponding MSI dictionary can be derived. Similarly, Zhu et al. [36] advocated for scene-complete dictionaries to facilitate sparse representation. In scenarios where the SRF is unknown, Fotiadou et al. [21] learnt the HSI and MSI dictionaries jointly by exploiting their relational constraints.

Despite their theoretical elegance, dictionary-based methods have fundamental limitations. First, their performance is constrained by the quality of the learned dictionary, often resulting in poor generalization to unseen scenes. Furthermore, their pixel-wise or patch-wise operation neglects the global spatial context and long-range dependencies that deep learning models excel at capturing. These inherent constraints consequently drove the field's transition to deep neural networks.

In recent years, deep learning has shown remarkable performance across a variety of vision tasks, owing to its powerful feature extraction capabilities [29], [37], [38], [39]. CNN-based methods offer clear optimization goals and stable training processes, allowing for flexible combination strategies [40], [41]. Shi et al. [26] utilized a residual connection network structure to enhance the generation effect. Koundinya et al. [42] proposed that 3-D convolutions perform better than 2-D convolutions in SSR tasks. Yan et al. [43] constructed U-Net-based network structures incorporating coordinate information to improve feature extraction. Despite the powerful nonlinear feature extraction capabilities of CNN-based methods, their limited receptive fields restrict the effective extraction of global information, leading to suboptimal generation results. GAN-based methods benefit from the generative adversarial process, which often leads to finer reconstructed images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:23:09 UTC from IEEE Xplore. Restrictions apply.

Alvarez-Gila et al. [30] incorporated contextual information into GANs to overcome the issue of localized information loss. Cao et al. [44] utilized conditional GANs to achieve bidirectional conversion between RGB and HSIs. GANs have also been successfully applied to the task of remote sensing image super-resolution, yielding remarkable results [45], [46]. GAN-based methods can achieve excellent generation results, but the adversarial learning process significantly increases training complexity and results in instability due to the influence of external factors.

B. Diffusion-Based Methods

The diffusion model operates via a bidirectional process. The forward process systematically corrupts input data by incrementally adding Gaussian noise, ultimately transforming the data into an isotropic Gaussian distribution. Conversely, the reverse process employs a learnable neural network to reconstruct the original data from this noise. Ho et al. [47] established a critical foundation for this field by proposing the denoising diffusion probabilistic model, which simplified the training objective and refined the model architecture. Building on this, Song et al. [48] introduced the denoising diffusion implicit model, which utilizes a non-Markovian sampling process to significantly reduce inference time and broaden the model's practical applicability. Rombach et al. [49] further advanced the field by shifting the diffusion process from the computationally expensive high-dimensional space to a low-dimensional latent space, enabling the efficient generation of high-resolution images.

These foundational models have inspired numerous valuable works in related downstream tasks. In image generation, for instance, Dhariwal and Nichol [31] improved the underlying U-Net architecture and introduced a classifier guidance technique to enhance the quality of generated images. To enable conditional generation, Hertz et al. [50] developed a method to alter image concepts by feeding feature maps into a cross-attention mechanism. The diffusion model has also demonstrated strong performance in super-resolution. Saharia et al. [33] achieved high-resolution results by conditioning the process on low-resolution images and applying iterative refinement. Wu et al. [51] pioneered the use of diffusion models in hyperspectral super-resolution, guiding the process with image feature maps. In the remote sensing domain, Xiao et al. [52] designed a lightweight noise prediction network, establishing an effective paradigm for such applications.

The application of diffusion models to SSR is still in its nascent stages. Early efforts in this domain include the work of Liu et al. [53], who employed a VQGAN to map HSIs into a latent space, thereby reducing sampling time and the number of model parameters. Separately, Chen et al. [54] adopted a cascaded method designed for the progressive learning of HSI spectral characteristics. While these approaches represent valuable contributions, their performance is fundamentally constrained by the limited availability of high-quality HSI data. To address this limitation, our work introduces an enhanced diffusion model tailored to the specific characteristics of the SSR task. We propose a dual-stage, single-sample optimization architecture designed to improve performance in limited-sample and even zero-shot scenarios. As a result, our proposed DPDM requires only a single MSI to reconstruct a complete HSI.

III. Method

We introduce a DPDM, as illustrated in Fig. 2, which consists of two key components: the PDM and the SRM. The PDM takes PIN as a condition and performs iterative processing within a single-sample optimization diffusion framework to recover an HSI. To address the limitations in the fine-grained spectral structure and spatial characteristics of the image recovered in the first stage, the SRM is employed. The SRM further enhances the spatial–spectral features, ultimately generating the final high-quality output.

A. Preliminary of the Diffusion Process

In the forward diffusion process, noise added as the variable $\mathbf{x_t} \in \mathbb{R}^{H\times W\times S}$ is sampled at time $t = 1, \ldots, T$, and the distribution of the original data is gradually destroyed. The forward process is a Markov chain, with its transitions described by the following conditional distributions:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\mathbf{x}_t;\ \sqrt{1-\beta_t}\cdot\mathbf{x}_{t-1}, \beta_t\cdot I\right)\ \ \forall t \in \{1,\ldots,T\} \quad (1)$$

where T is the total number of steps in the forward process, $\beta_1, \cdots, \beta_T$ are hyperparameters representing the variance schedule in the forward process, I is the identity matrix with the same dimension as $\mathbf{x}_0 \in \mathbb{R}^{H\times W\times S}$, and $\mathcal{N}(\mathbf{x}; \mu, \sigma)$ is a normal distribution with mean μ and variance σ.

In the reverse process of the diffusion model, the goal is essentially to perform reverse modeling of the distribution $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$. This modeling process can be learned by employing a neural network parameterized by θ_t as follows:

$$p_{\theta_t}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\mathbf{x}_{t-1}; \mu_{\theta_t}(\mathbf{x}_t, t), \sigma_{\theta_t}(\mathbf{x}_t, t)\right) \quad (2)$$

where μ_{θ_t} and σ_{θ_t} denote the mean and covariance of the state $\mathbf{x}_{t-1}$, respectively.

Our training object is the distribution of $p_{\theta_t}(\mathbf{x}_0)$, and we need to obtain its log-likelihood function. We minimize this problem by introducing Kullback–Leibler (KL) divergence to construct a negative log-likelihood upper bound for the training objective

$$\begin{aligned} -\log p_{\theta_t}(\mathbf{x}_0) \le & -\log p_{\theta_t}(\mathbf{x}_0) \\ & + D_{KL}\left(q\left(\mathbf{x}_{1:T}|\mathbf{x}_0\right)\|p_{\theta_t}\left(\mathbf{x}_{1:T}|\mathbf{x}_0\right)\right) \end{aligned} \quad (3)$$

then we can optimize the variational lower bound

$$\begin{aligned} & \mathbb{E}_{q(\mathbf{x}_{0:T})}\left[\log q(\mathbf{x}_{1:T}|\mathbf{x}_0) - \log p_{\theta_t}(\mathbf{x}_{0:T})\right] \\ & \le -\mathbb{E}_{q(\mathbf{x}_0)} log p_{\theta_t}(\mathbf{x}_0). \end{aligned} \quad (4)$$

We inference the relation between $\mathbf{x}_t$ and $\mathbf{x}_0$ by re-parameterizing (1) as follows:

$$\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_0 + (1-\alpha_t)\epsilon_x \quad (5)$$

where $\epsilon_x \in \mathbb{R}^{H\times W\times S}$ is the Gaussian noise.

For the expression of the distribution $p_\theta(\mathbf{x}_0)$, it can be obtained by combining (4) and (5) as

$$\arg\min_{\theta_t} \left|\left|\mathbf{x}_t - \left(\sqrt{\alpha_t}\mathbf{x}_0 + \sqrt{1-\alpha_t}\epsilon_x\right)\right|\right|^2. \quad (6)$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:23:09 UTC from IEEE Xplore. Restrictions apply.

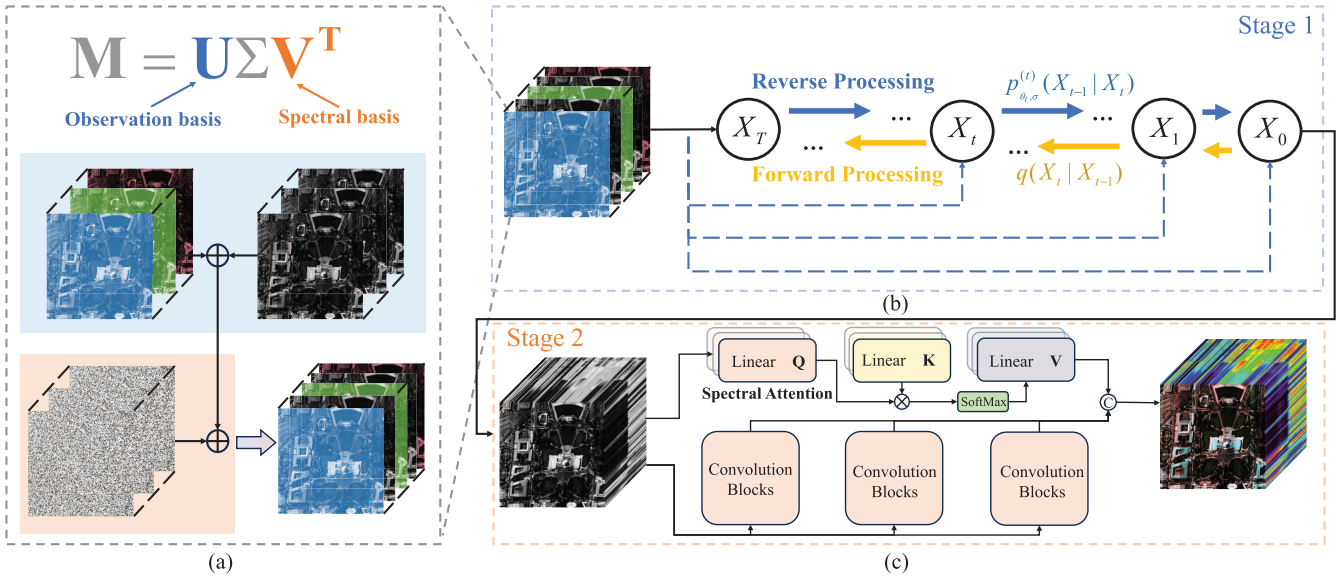


Fig. 2. Diagram of the proposed DPDM architecture. (a) PIN construction process. We construct PIN in two sequential stages: first, we project the MSI and the corresponding differential image onto the observation basis and subsequently reconstruct the initial HSI using the resulting singular values and spectral basis. (b) PDM pipeline. The PDM uses a single-sample optimization noise prediction network in which most of the skip connections are removed to strengthen the network's own prior and delay overfitting. (c) SRM composition. The SRM can be divided into three levels, each consisting of convolution blocks and spectral attention blocks.

B. Prior-Driven Diffusion Module

HSIs are characterized by complex spectral structures and limited sample availability, both of which significantly constrain the performance of SSR tasks. To address these limitations, we propose the PDM that integrates both data priors and network structure priors to facilitate high-quality HSI generation. Specifically, we develop a prior-driven sampling mechanism guided by a construction prior $\bar{\mathbf{c}}$, which leverages the spectral information inherent in multispectral observations. Throughout the sampling process, this construction prior $\bar{\mathbf{c}}$ enables the progressive transfer of spectral features across sampling iterations, thereby enhancing the fidelity of reconstructed spectral details.

Furthermore, the noise prediction network within the diffusion framework is restructured to more effectively model the intricate spectral dependencies present in HSIs. The noise prediction network adopts a U-shaped architecture. Instead of convolutional layers, we employ a lightweight spectral attention mechanism to enhance the network's ability to globally represent spectral features across scales. In the noise prediction network, nearly all skip connections are removed to encourage the denoising process to prioritize the network's intrinsic structural priors over data-driven dependencies [55].

We describe the sampling process more concisely using mathematics. The spectral response matrix $\mathbf{M} \in \mathbb{R}^{s\times S}$ can be decomposed into as follows:

$$\mathbf{M} = \mathbf{U}\Sigma\mathbf{V}^{\mathbf{T}} \tag{7}$$

where $\mathbf{U} \in \mathbb{R}^{s\times s}$ and $\mathbf{V} \in \mathbb{R}^{S\times S}$ are orthogonal matrices and $\Sigma \in \mathbb{R}^{s\times S}$ is a matrix of singular values in a diagonal structure obtained by decomposing $\mathbf{M}$, including $s_1 \geq s_2 \geq \cdots \geq s_n$.

The module input $\mathbf{x}_T$ is a complex high-dimensional noise, making recover of $\mathbf{x}_0$ extremely challenging. Therefore, we introduced the prior $\bar{\mathbf{c}}$ to assist the process. PIN $\bar{\mathbf{c}}$ is constructed as follows:

$$\bar{\mathbf{c}} = \Sigma^{\dagger}\mathbf{U}^{\mathbf{T}}\left(\hat{\mathbf{y}} + \vartheta\epsilon_y\right) \tag{8}$$

where $\epsilon_y \in \mathbb{R}^{H\times W\times s}$ is the noise, ϑ is the noise weight, $\dagger$ represents the Moore–Penrose generalized inverse, and $\mathbf{y} \in \mathbb{R}^{H\times W\times s}$ denotes the MSI, which corresponds to an RGB image in the following text. $\hat{\mathbf{y}}$ can be obtained by $\hat{\mathbf{y}} = \mathbf{y} + \triangle\mathbf{y}$. The component $\triangle\mathbf{y}$ is obtained by differentiating $\mathbf{y}$ with respect to its bands.

We seek to embed the constructed prior into the variables of the reverse diffusion process, with the goal of improving the quality of hyperspectral data obtained through final sampling. We define the spectral space of high-dimensional data as follows: $\bar{\mathbf{c}}^{(i)}$ is the ith index of the $\bar{\mathbf{c}}$ and $\bar{\mathbf{x}}^{(i)}$ is the ith index of the $\bar{\mathbf{x}}_t = \mathbf{V}^T\mathbf{x}_t$. For each index i in $\bar{x}_t$, the variational distribution can be defined as

$$p^{(T)}_{\theta_T,\sigma}\left(\bar{\mathbf{x}}^{(i)}_T|\mathbf{c}\right) = \begin{cases}\mathcal{N}\left(\bar{\mathbf{c}}^{(i)}, \sigma_T^2 - \dfrac{\sigma_{\mathbf{c}}^2}{s_i^2}\right), & \text{if } s_i > 0\\ \mathcal{N}\left(0, \sigma_T^2\right), & \text{if } s_i = 0\end{cases} \tag{9}$$

$$p^{(t)}_{\theta_t,\sigma}\left(\bar{\mathbf{x}}^{(i)}_t|\mathbf{x}_{t+1}, \mathbf{c}\right) = \begin{cases}\mathcal{N}\left(\bar{\mathbf{x}}^{(i)}_{\theta_t,t} + \sqrt{1-\eta^2}\sigma_t\dfrac{\bar{\mathbf{x}}^{(i)}_{t+1} - \bar{\mathbf{x}}^{(i)}_{\theta_t,t}}{\sigma_{t+1}}, \eta^2\sigma_t^2\right), & \text{if } s_i = 0\\ \mathcal{N}\left(\bar{\mathbf{x}}^{(i)}_{\theta_t,t} + \sqrt{1-\eta^2}\sigma_t\dfrac{\bar{\mathbf{c}}^{(i)} - \bar{\mathbf{x}}^{(i)}_{\theta_t,t}}{\sigma_c s_i}, \eta^2\sigma_t^2\right), & \text{if } \sigma_t < \frac{\sigma_c}{s_i}\\ \mathcal{N}\left((1-\eta_b)\bar{\mathbf{x}}^{(i)}_{\theta_t,t} + \eta_b\bar{\mathbf{c}}^{(i)}, \sigma_t^2 - \dfrac{\sigma_c^2}{s_i^2}\eta_b^2\right), & \text{if } \sigma_t \geq \frac{\sigma_c}{s_i}\end{cases} \tag{10}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:23:09 UTC from IEEE Xplore. Restrictions apply.

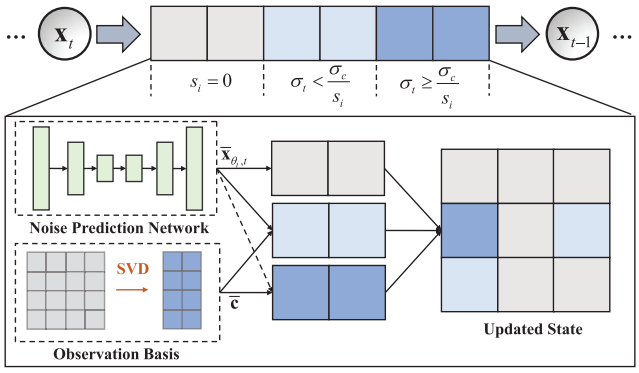


Fig. 3. Schematic of a single denoising step of the PDM. During the reverse sampling process, the PDM uses singular values and noise levels as criteria to partition the image into three parts.

where σ_t depends on the variance of $\mathbf{x}_t$ and η and η_b are hyperparameters that can control the noise level inserted in each step. $\mathbf{x}_{\theta_t,t}$ can be obtained by sampling, and $\bar{\mathbf{x}}_{\theta_t,t}$ can be obtained after left multiplication by $\mathbf{V}$. The relevant parameters θ_t involved in the model are all generated by random initialization.

Equations (8) and (9) detail the construction of PIN, where pure noise is not directly used as the input to the diffusion module. Instead, we leverage the spectral basis $\mathbf{V}$ and the observation basis $\mathbf{U}$ to construct a structured, high-dimensional input for the PDM. Specifically, the process begins by employing the observation basis $\mathbf{U}$ to analyze the spectral characteristics of the MSI and its per-band difference image. The resulting coefficients are then weighted by the singular value matrix $\mathbf{\Sigma}$ to perform the low-to-high-dimensional mapping. This allows the spectral basis $\mathbf{V}$ to reconstruct the features in the target high-dimensional space.

As illustrated in Fig. 3 and detailed in (10), the PDM partitions the image into three components based on singular values s_i and the signal strength σ_i. For components with a zero-singular value, indicating a complete absence of prior information, the output is generated entirely by the noise prediction network. When the singular value is nonzero and σ_t is below the threshold σ_c/s_i, the corresponding component is considered to contain a mix of prior and unknown information; its output is, therefore, a combination of the prior $\bar{\mathbf{c}}^{(i)}$ and the output from the noise prediction network. Finally, when a component has a nonzero singular value and its signal strength σ_t is greater than the threshold σ_c/s_i, it exhibits a high degree of similarity to the prior. The output is thus generated predominantly from the prior $\bar{\mathbf{c}}^{(i)}$, with a smaller contribution from the noise prediction network.

C. Spectral Refinement Module

The noise introduced in the diffusion process will destroy the spatial and spectral structure of the HSIs. As shown in Fig. 4, there is a substantial difference between remote sensing images and natural images with respect to spatial structure and texture details. Consequently, noise-induced degradation is often more pronounced in remote sensing images. Thus, we design the SRM to overcome the effect of noise introduction.

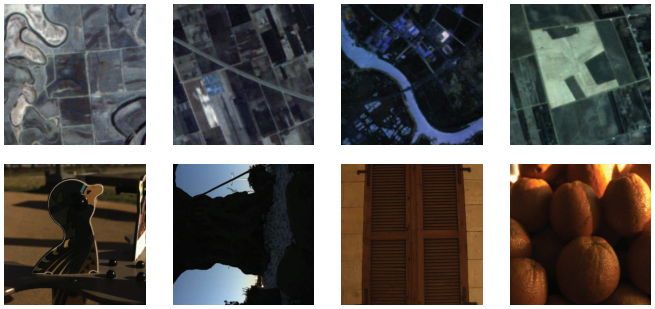


Fig. 4. Remote sensing RGB image of test dataset (top row) and natural RGB images (bottom row).

The specific composition of the SRM architecture is shown in Fig. 5.

The HSI generated in the first stage can be defined as $\mathbf{I} \in \mathbb{R}^{H\times W\times S}$. It is processed by pixelunshuffle in the SRM to obtain a multiscale feature map, which can be represented as

$$\mathbf{I}' = \mathcal{D}_r(\mathbf{I}) \tag{11}$$

where $\mathbf{I}' \in \mathbb{R}^{H/r,W/r,S\times r^2}$, r is the downscale factor, and $\mathcal{D}_r$ denotes pixelunshuffle. Afterward, $\mathbf{I}'$ will be fed into two branches, the convolutional branch and the spectral attention branch, respectively.

Among the convolutional branches, the procedure of $\mathbf{I}'$ can be expressed as follows:

$$\mathbf{I}_{\text{Conv}} = \text{Concat}\left(f_{\text{Conv1}}(\mathbf{I}'), f_{\text{Conv3}}(\mathbf{I}')\right) \tag{12}$$

where Conv1 and Conv3 refer to convolution 1×1 and convolution 3×3. Subsequently, the feature maps being concatenated will be re-aggregated as

$$\mathbf{I}'_{\text{Conv}} = f_{\text{Conv3}}\left(\text{LeakyReLU}(\mathbf{I}_{\text{Conv}})\right). \tag{13}$$

In the attention branch, the input $\mathbf{I}'$ will first be converted to $\mathbf{I}_{\text{in}} \in \mathbb{R}^{HW\times S}$ and is then used for attention calculation as follows:

$$\mathbf{Q} = \mathbf{W_q}\mathbf{I}_{\text{in}},\ \mathbf{K} = \mathbf{W_k}\mathbf{I}_{\text{in}},\ \mathbf{V} = \mathbf{W_v}\mathbf{I}_{\text{in}} \tag{14}$$

where $\mathbf{W_q}, \mathbf{W_k}, \mathbf{W_v} \in \mathbb{R}^{S\times S}$. We refer to [29], split $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ into N heads by the number of spectral bands: $\mathbf{Q} = [\mathbf{Q}_1, \mathbf{Q}_2, \mathbf{Q}_3 \cdots, \mathbf{Q}_N]$, $\mathbf{K}$ and $\mathbf{V}$ have the same dimension. Each head has a dimension of $h_d = S/N$.

The computation of attention can be expressed as follows:

$$\text{head}_l = \mathbf{V}_l \text{Softmax}\left(\mathbf{K}_l^T \mathbf{Q}\right). \tag{15}$$

Subsequently, the computational process of the constructed multihead attention can be expressed as follows:

$$I'_{\text{attention}} = \text{LN}\left([\text{head}_1; \text{head}_2, \cdots, \text{head}_N]\,\mathbf{W^o} + f_{\text{pos}}(\mathbf{V})\right) \tag{16}$$

where LN is the LayerNorm activation function, $\mathbf{W^o}$ are learnable parameters, and f_{pos} is the position embedding function. Subsequently, the feature maps from the two branches are combined.

A cascade refinement strategy is employed to enhance the spectral refinement capabilities of the SRM. This approach is inspired by the principles of deep image prior (DIP) [55] and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:23:09 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Generation Process of the DPDM

Input: multispectral image $\mathbf{y}$, spectral response matrix $\mathbf{M}$

1: **Initialization:** random sample $\mathbf{x}_T \sim \mathcal{N}(0, \delta^2)$, f_{SRM} is the SRM, ϵ_θ is the single-sample noise predictor, ϵ is the noise.
2: Initialize the weights of the whole network;
3: Construct $\mathbf{c} = \mathbf{y} + \vartheta\epsilon_y$
4: Obtain $\bar{\mathbf{x}}_T$ via Eq.(9);
5: **for** $t = T : 1$ **do**
6: $\quad \bar{\mathbf{x}}_{\theta_t,t} = \epsilon_\theta(\bar{\mathbf{x}}_t)$;
7: $\quad$ Obtain $\hat{\mathbf{x}}_{t-1}$ via reparameterizing Eqn.(10);
8: **end for**
9: $\tilde{\mathbf{x}}_{\text{HSI}} = f_{\text{SRM}}(\hat{\mathbf{x}}_0)$;

Output: generated $\tilde{\mathbf{x}}_{\text{HSI}}$.

operates on a single-sample optimization basis. Specifically, the output from a preceding stage serves as both the input and the reconstruction target for the subsequent SRM stage. The optimization is conducted independently of the ground-truth HSI, relying on the network's inherent structure before refining the spectral features. To mitigate overfitting and preserve spectral details during the iterative process, we employ an early stopping strategy analogous to that used in DIP. For a comprehensive understanding, the overall DPDM process is detailed in Algorithm 1.

IV. Experiments

In this section, we conduct extensive comparative experiments to evaluate our method against other state-of-the-art approaches for the SSR task. Moreover, we perform meticulous ablation studies on the DPDM to verify the effectiveness of each module and its parameter settings.

A. Dataset and Evaluation Indicators

We use remote sensing HSIs from Chinese Orbita hyperspectral satellites (OHS) to construct the training dataset. The spatial resolution of the dataset is 10 m, and the spectral band ranges from 400 to 1000 nm with an increment of 2.5 nm. We segmented the original data and selected 9025 HSIs of size 256×256 and sampled them into 31 bands from the original bands. During the training process, we randomly partition the available image pairs into a training set (7581 pairs), a validation set (722 pairs), and a test set (722 pairs).

To evaluate the generalization ability of the model, we test different methods using the public dataset Washington DC Mall. The dataset covers a spectral range of 400–2500 nm with 191 bands. The dataset captures an urban scene in Washington DC, USA, with a spatial resolution of 3 m and an image size of 307×1280 pixels. During the processing, we divided the original pixels 307×1280 based on the size of 256×256, resulting in the creation of five test images. In particular, the bands and wavelengths of the Washington DC Mall dataset significantly exceed those of the training dataset. To harmonize the spectral characteristics of the test data with our training domain, we first select a comparable spectral range and then uniformly resample the bands to a count of 31.

The Pavia Center dataset spans a spectral range of 430–860 nm with 102 bands. Captured by the ROSIS sensor over an urban area in Pavia, Italy, it has a spatial resolution of 1.3 m and an image size of 1096×715 pixels. During processing, the original image 1096×715 pixels is divided into patches of size 256×256, resulting in 8 test images. For this dataset, we employ the same band processing method used for the Washington DC Mall dataset.

In our experiments, we adopt the spectral response matrix (cie_1964_w_gain) from [56]. The multispectral data are generated by multiplying this spectral response matrix with the corresponding hyperspectral data. To resolve inconsistencies in wavelength overlap between the datasets and the response matrix, linear interpolation is applied to the OHS, Washington DC Mall, and Pavia Center datasets to achieve proper band alignment.

To compare the performance between different algorithms fairly, we evaluate the performance of the SSR task using the following four evaluation criteria: the mean relative absolute error (MRAE), the root-mean-square error (RMSE), the spectral angle mapper (SAM), and the peak signal-to-noise ratio (PSNR). The MRAE and RMSE quantify the relative and absolute errors in the spatial structure between the reconstructed and real HSIs, with smaller values indicating lower errors. The SAM measures the spectral similarity between the two HSIs. Lower SAM values indicate a closer alignment of their spectral structures, which results in higher spectral fidelity in the generated HSIs. The PSNR assesses the preservation of spatial details, where larger values denote more accurate restoration of the spatial structure.

B. Implementation Details

1) Parameter Configuration: In the PDM, the total number of diffusion steps is set to 1000, with no gradient backpropagation performed during the first 500 steps. The random seed is fixed at 3407. The initial learning rate for the first stage is set to 1×10^{-4}, using the Adam optimizer. The SRM underwent training for 500 iterations using a cosine learning rate scheduler that progressively decreased the rate from 1×10^{-4} to 1×10^{-5}. The Adam optimizer is employed with its default parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$.

2) Experimental Details: Throughout the entire training process, comparative methods are trained exclusively on the OHS training dataset. All experiments are implemented on PyTorch with a 24 GB memory NVIDIA RTX 3090 GPU.

C. Comparison With State-of-the-Art Methods

To verify the effectiveness of the proposed DPDM method, we conduct both qualitative and quantitative comparisons with other state-of-the-art methods, including a sparse coding-based method Arad [19], CNN-based methods EDSR [57], AWAN [58], HSRNet [59], HSCNN+ [26], HRNet [28], SIGNet [60], MPRNet [61], HINet [62], AGD-Net [63], a GAN-based method Adv-GAN [30], a Transformer-based method Mst++ [29], and a diffusion-based method SCDM [54].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:23:09 UTC from IEEE Xplore. Restrictions apply.

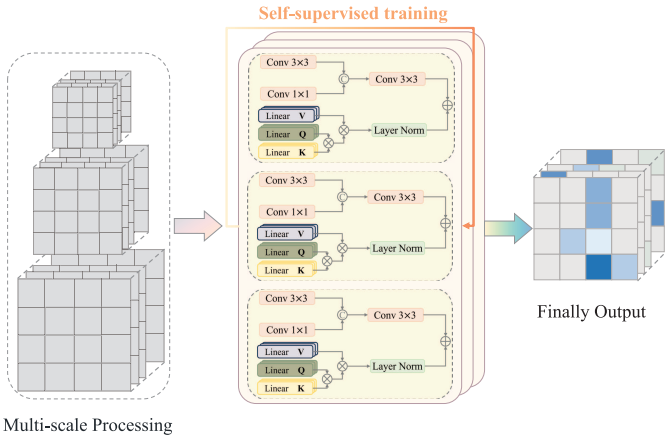


Fig. 5. Schematicof the SRM comprises three layers, each consisting of a convolutional block and a spectral attention block. The convolutional block is responsible for extracting spatial features, whereas the spectral attention block is designed to capture spectral representations.

In the model training phase, we train EDSR, AWAN, HSRNet, HSCNN+, HRNet, MPRNet, HINet, AGD-Net, Adv-Net, Mst++, and SCDM on the OHS dataset. This is because the weight files for the methods listed above are trained on natural images or are unavailable, which could lead to unequal comparisons. In contrast, Arad can be directly inferred without training. The SIGNet uses the prior from the specified sensor's SRF during training, which makes it difficult to train it on the OHS dataset. Therefore, we use the weight provided by the authors.

1) Qualitative Evaluation: Fig. 6 presents the true-color reconstruction results for the compared methods on the OHS dataset. The proposed DPDM exhibits superior overall visual quality, with its results most closely matching the ground-truth (GT) image in terms of color fidelity and spatial structural integrity. This superiority is particularly evident in the detailed enlargements, where the DPDM accurately restores the texture of the farmland [Fig. 6(a)] and the contours of the road [Fig. 6(c)]. Among the competing methods, the Mst++, MPRNet, and HRNet also perform commendably in restoring textural details, yet they exhibit slight color deviations when compared to the GT. In contrast, other methods suffer from significant visual defects. The results from the SIGNet and SCDM are corrupted by severe noise and artifacts, which obscure discernible details. Similarly, Arad and EDSR exhibit noticeable color distortion and edge blurring. To facilitate a more detailed comparison of reconstruction accuracy, error maps are generated to visualize the pixel-wise differences between the reconstructed HSI and the GT.

Fig. 7 illustrates the generation errors of various comparison methods across different bands of HSIs. To ensure comprehensive comparison results, we conducted comparative analysis using three distinct spectral channels: band 5 (visible), band 15 (visible to near-infrared transition), and band 31 (near-infrared). The error graph provides an intuitive assessment of the SSR performance of the methods under comparison. Band 5, in the blueish-green visible spectrum, presents challenges for spectral detail restoration due to atmospheric scattering effects. In our visual analysis, all methods except the SCDM demonstrate high spectral accuracy in this region. Band 15, at the near-infrared spectrum's edge, serves as a transition between visible and near-infrared bands and effectively indicates vegetation spectral fidelity. Comparative analysis reveals that the SCDM, SIGNet, EDSR, and AWAN exhibit distorted spectral reconstruction, particularly in the vegetation regions of test image 2. Band 31, in the near-infrared region, typically contains spatial structure information of vegetation and water bodies. The error of this band demonstrates that only the DPDM and Adv-GAN achieve satisfactory reconstruction of water bodies (upper left of test image 1) and vegetation (test image 3 and bottom of test image 2). The DPDM maintains consistently low reconstruction error across nearly all bands, indicating superior spectral fidelity throughout the visible to near-infrared spectrum.

Fig. 8 illustrates the reconstruction results of various SSR methods. A comparison with the GT reveals that the proposed DPDM exhibits exceptional restoration performance in both spatial structure details and spectral fidelity. In test image 1 (band 31) and test image 2 (band 31), the DPDM outperforms the other methods by avoiding significant blurring or loss of detail, particularly in regions with complex spatial structures or at the edges. In test image 3 (band 27) and test image 4 (band 17), the DPDM demonstrates a strong match with the reconstructed spectrum. A comprehensive analysis of Figs. 7 and 8 suggests that achieving high spectral reconstruction accuracy in the near-infrared band (750–1000 nm) is challenging for various reconstruction methods. Additionally, clearly restoring the spatial structure in the visible band is also difficult, particularly as demonstrated by the 17-band test image 4 in Fig. 8.

As shown in Fig. 9, we plot the full-band index graphs for all comparison methods. Fig. 9(c) supports our analysis of spectral reconstruction accuracy. Most methods maintain low spectral errors in the visible light band (400–750 nm) but exhibit significant errors in the near-infrared band above 750 nm. This may be due to the input image containing only visible light information, with minimal near-infrared data. As shown in Fig. 9(a) and (b), the accuracy of spatial structure recovery in the visible light is low, which contradicts intuition. We think this may result from the conflict between the spatial anisotropy of remote sensing images and the inherent translation invariance of the convolution module. Consequently, the decrease in spatial indicator accuracy is mitigated in the Transformer-based method Mst++. Additionally, the DPDM demonstrates outstanding performance in both spatial structure restoration and spectral accuracy preservation, confirming the effectiveness of our proposed method.

We compare the spectral curves of key objects in the generated HSIs using different methods in Fig. 10. In the figure, the DPDM demonstrates high spectral reconstruction accuracy, particularly in challenging scenes. For example, Fig. 10(b) and (e) includes vegetation, which produces an increasing trend in the near-infrared band (900 nm). Notably, the spectral reflectance generated by the DPDM varies adaptively in response to spectral within complex scenes. This adaptability highlights the method's robustness in handling conditions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:23:09 UTC from IEEE Xplore. Restrictions apply.

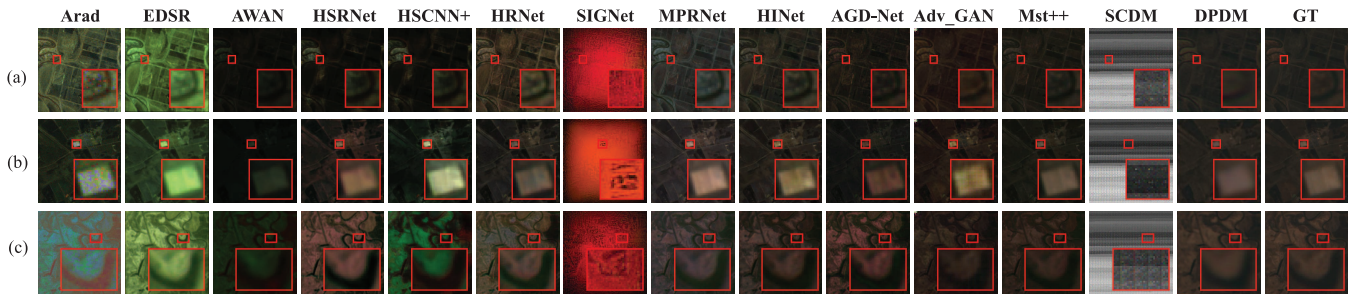


Fig. 6. Visual comparison of true color (RGB) images synthesized from the OHS dataset. The RGB images are generated by mapping spectral bands 14, 7, and 2 to the red, green, and blue channels, respectively. (a) Agricultural area. (b) Built-up area. (c) River delta.

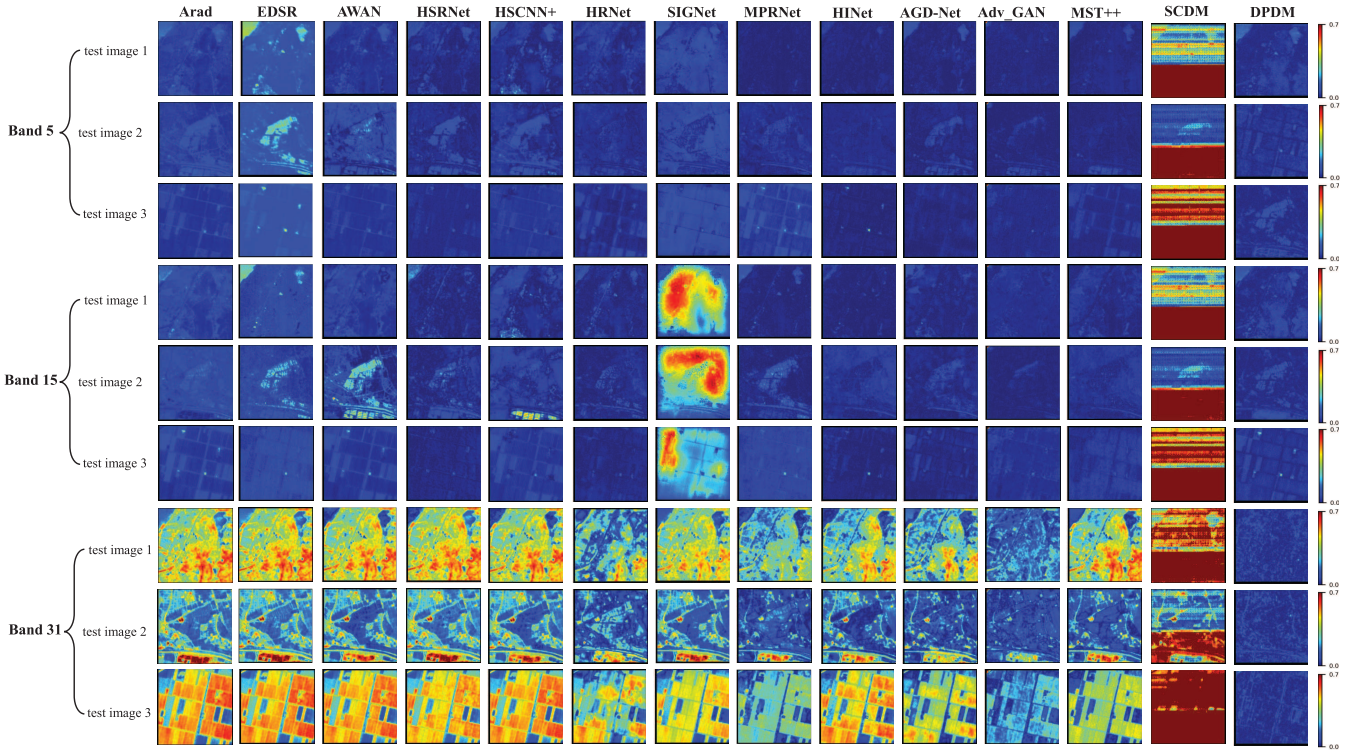


Fig. 7. Absolute error maps between reconstructed HSIs and GT for bands 5, 15, and 31 on the OHS dataset. The color bar denotes the error range.

Finally, to evaluate generalization performance, the reconstruction errors for all methods are analyzed on the public Washington DC Mall and Pavia Center datasets. As depicted in the error maps in Fig. 11, the DPDM achieves exceptionally low reconstruction error across both datasets, demonstrating its robust generalization capability. The GAN-based Adv-GAN and the spectral dictionary-based Arad delivered the next-best performance, also indicating strong generalization. In contrast, numerous CNN-based methods (e.g., EDSR, AWAN, HSRNet, HSCNN+, and MPRNet) exhibit errors that are primarily concentrated in regions with high-frequency information, such as building edges, road boundaries, and complex textures. This suggests that while these models can capture the global image structure, they are limited in their ability to recover fine-grained details. Lastly, the results for the SCDM are characterized by horizontal stripe artifacts, which likely indicate a systemic flaw introduced by its generative paradigm.

TABLE I

AVERAGE QUANTITATIVE METRICS OF THE COMPARISON METHOD ON THE OHS DATASET

	Method	MRAE ↓	RMSE ↓	SAM ↓	PSNR ↑
Sparse Coding-based	Arad [19]	1.6695	0.1488	0.4155	16.0998
CNN-based	EDSR [57]	0.9949	0.1501	0.9427	15.0062
	AWAN [58]	0.6897	0.1228	0.5572	16.8685
	HSRNet [59]	0.6514	0.1125	0.7105	16.9092
	HSCNN+ [26]	0.6209	0.1137	0.5343	17.4877
	HRNet [28]	0.6841	0.1004	0.3726	18.0719
	SIGNet [60]	0.7552	0.1655	0.6264	13.2592
	MPRNet [61]	0.6544	0.0865	0.2882	19.6189
	HINet [62]	0.5233	0.0864	0.3607	19.7686
	AGD-Net [63]	0.4436	0.0775	0.3480	20.6566
GAN-based	Adv-GAN [30]	0.4887	0.0629	**0.2291**	21.1482
Transformer-based	Mst++ [29]	0.4642	0.0780	0.3216	20.9210
Diffusion-based	SCDM [54]	0.7858	0.6759	0.5165	12.4090
	DPDM	**0.3323**	**0.0392**	0.2386	**25.6142**

2) Quantitative Evaluation: Table I presents the average quality evaluation results of all SSR methods on the OHS

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:23:09 UTC from IEEE Xplore. Restrictions apply.

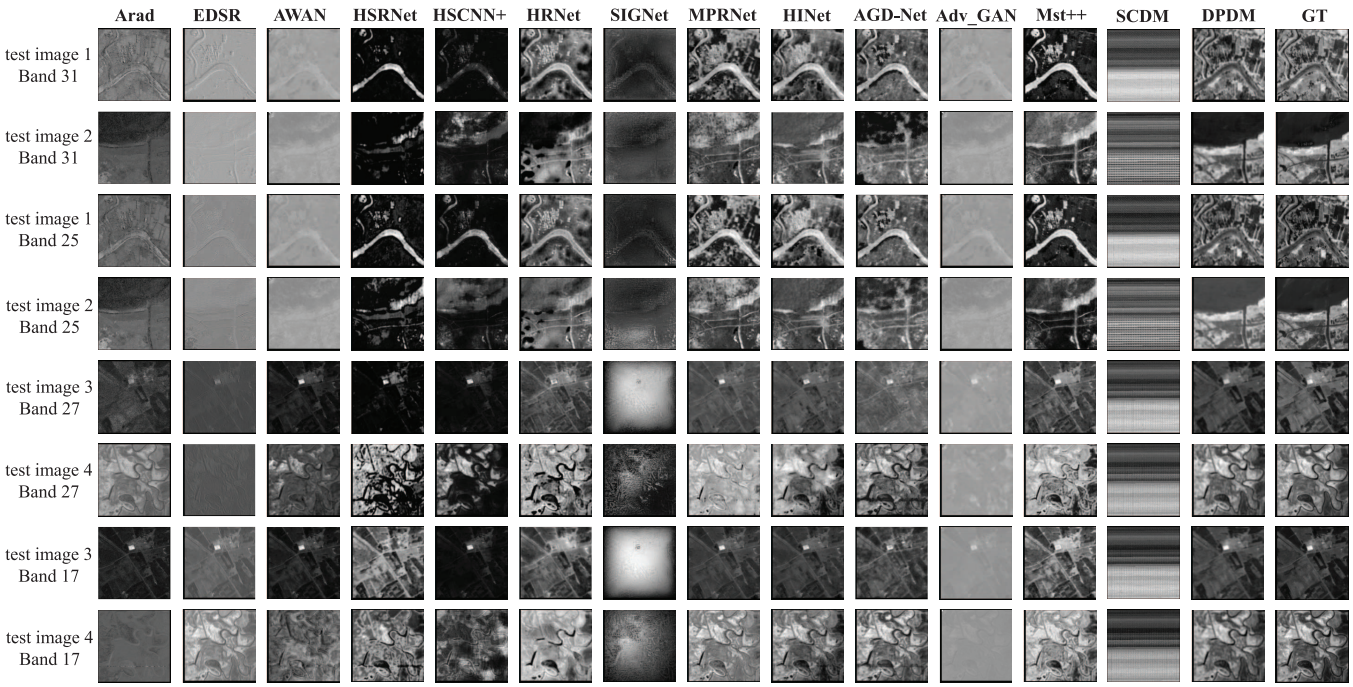


Fig. 8. Comparison charts of HSI bands generated by different methods.

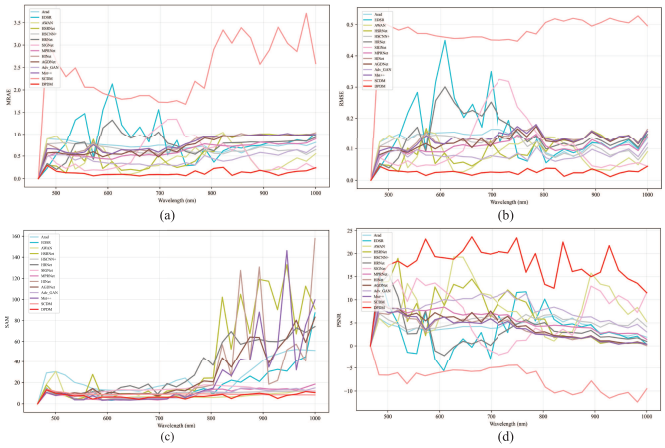


Fig. 9. Schematic of the band-wise comparison of different methods across 4 evaluation indicators. (a) MRAE. (b) RMSE. (c) SAM. (d) PSNR.

dataset. The best results are highlighted in bold, and the second-best results are underlined. The table shows that our method outperforms others in the spatial evaluation metrics MRAE, RMSE, and PSNR, while yielding suboptimal performance in the spectral fidelity metric SAM. This limitation may stem from the model's insufficient capacity to handle complex spectral present in certain scenes within the OHS dataset. The results indicate that the proposed method effectively reconstructs the spatial structure of the original HSI while preserving high spectral fidelity in most scenarios.

We evaluate the generalization performance of the model through cross-dataset experiments on the Washington DC Mall dataset, as shown in Table II. The experiments demonstrate that our method achieves the best results across all evaluation metrics, highlighting its superior generalization ability.

TABLE II

AVERAGE QUANTITATIVE METRICS OF THE COMPARISON METHOD ON THE WASHINGTON DC MALL DATASET

	Method	MRAE ↓	RMSE ↓	SAM ↓	PSNR ↑
Sparse Coding-based	Arad [19]	0.4201	0.1122	0.2475	19.4926
CNN-based	EDSR [57]	0.4493	0.1538	0.4504	14.7257
	AWAN [58]	0.5272	0.1788	0.3191	14.3278
	HSRNet [59]	0.6859	0.1816	0.4337	12.6548
	HSCNN+ [26]	0.5925	0.2076	0.2896	13.0822
	HRNet [28]	0.4365	0.1537	0.2969	15.4795
	SIGNet [60]	0.8024	0.2174	0.5555	12.4374
	MPRNet [61]	0.5015	0.1377	0.3733	15.7222
	HINet [62]	0.3895	0.1151	0.3099	17.3981
	AGD-Net [63]	0.5351	0.1121	0.2334	18.0785
GAN-based	Adv-GAN [30]	<u>0.0776</u>	<u>0.0424</u>	<u>0.0741</u>	<u>24.4425</u>
Transformer-based	Mst++ [29]	0.5386	0.1270	0.3539	16.0776
Diffusion-based	SCDM [54]	0.3331	0.2111	0.3071	13.5640
	DPDM	**0.0402**	**0.0229**	**0.0387**	**29.6054**

Notably, the performance of sparse coding-based, CNN-based, and Transformer-based methods has declined in the Washington DC Mall dataset, while the generative methods Adv-GAN and DPDM show improvement. This indicates that the distribution fitting characteristics of the generative model contribute to enhanced generalization. Furthermore, a core advantage of the DPDM is its architectural design, which generates an HSI based solely on a single input sample. This approach eliminates reliance on external training datasets, thereby avoiding the generalization errors that can arise from training set bias. This independence renders the model highly suitable for scenarios requiring strong generalization or operating with limited data.

In addition, we analyze the performance of comparison methods on the Pavia Center dataset. The experimental results presented in Table III indicate that almost all methods exhibit

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:23:09 UTC from IEEE Xplore. Restrictions apply.

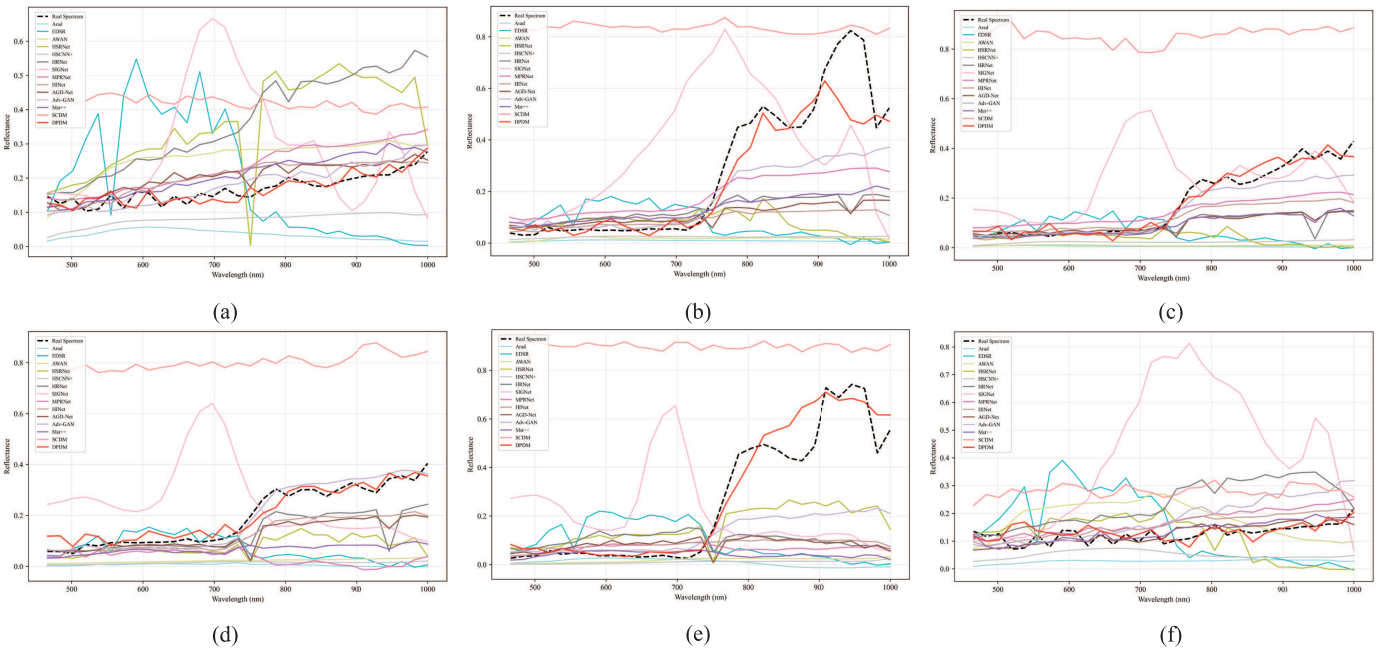


Fig. 10. Comparison of spectral curves for different methods across six types of ground objects. (a) Bare land. (b) Farmland. (c) Building. (d) Water body. (e) Forest. (f) Road.

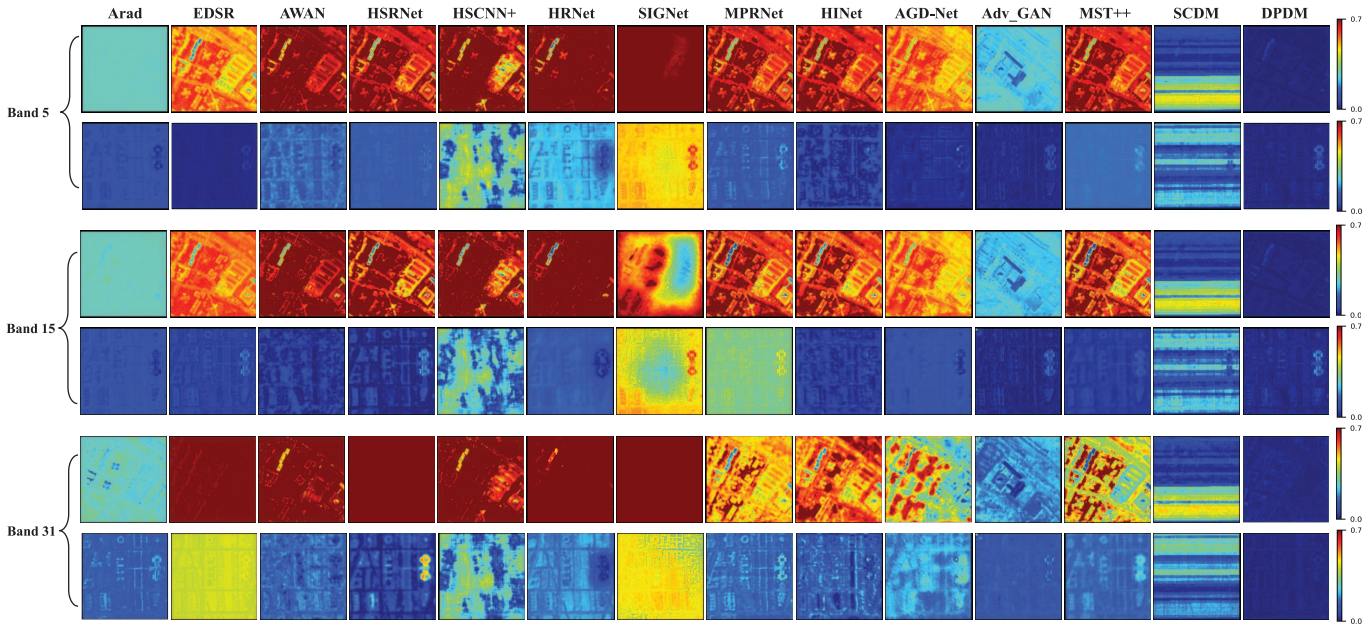


Fig. 11. Absolute error maps between reconstructed HSIs and GT for bands 5, 15, and 31 on the Pavia Center (Top) and Washington DC Mall (Bottom) datasets. The color bar denotes the error range.

declines in their performance indicators, except the DPDM. This phenomenon may be attributed to the fact that the Pavia Center dataset omits certain bands affected by noise during its construction, resulting in a discontinuous wavelength distribution. In contrast, the DPDM maintains its leading performance across all metrics. This success stems from its single-sample optimization architecture. By operating without a training dataset, the DPDM inherently avoids the performance degradation typically caused by distribution shifts between training and test data, thus demonstrating robust generalization capabilities.

Although the DPDM achieves competitive quantitative results on the three datasets, the algorithm's efficiency is affected by both the iterative optimization process inherent to diffusion models and the model's complexity. To objectively evaluate the efficiency of the DPDM, we, therefore, analyzed these metrics, with the results presented in Table IV. The iterative refinement architecture inherent to diffusion models

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:23:09 UTC from IEEE Xplore. Restrictions apply.

TABLE III
AVERAGE QUANTITATIVE METRICS OF THE COMPARISON METHOD ON THE PAVIA CENTER DATASET

	Method	MRAE ↓	RMSE ↓	SAM ↓	PSNR ↑
Sparse Coding-based	Arad [19]	0.7878	0.2275	0.1094	14.5568
CNN-based	EDSR [57]	0.7775	0.2208	0.3436	15.6839
	AWAN [58]	0.8741	0.2641	0.2455	13.8377
	HSRNet [59]	0.8523	0.2624	0.4282	13.2782
	HSCNN+ [26]	0.8999	0.2932	0.3091	12.4536
	HRNet [28]	0.8935	0.2883	0.3485	12.7433
	SIGNet [60]	0.7478	0.2448	0.4110	12.3600
	MPRNet [61]	0.7342	0.1894	0.2388	16.8763
	HINet [62]	0.7420	0.2071	0.1995	16.3325
	AGD-Net [63]	0.6211	0.2412	0.1548	13.1400
GAN-based	Adv-GAN [30]	0.7592	0.2072	0.2114	15.8261
Transformer-based	Mst++ [29]	0.6951	0.2019	0.2845	14.4920
Diffusion-based	SCDM [54]	0.1970	0.2232	0.2525	13.1479
	DPDM	**0.0175**	**0.0147**	**0.0189**	**32.3212**

TABLE IV
COMPARISON OF MODEL COMPLEXITY AND EFFICIENCY

	Method	Params (M)	Training Time(h)	Inference Time(s)
Sparse Coding-based	Arad [19]	-	-	-
CNN-based	EDSR [57]	2.42	44.2	0.08
	AWAN [58]	4.04	155.90	0.19
	HSRNet [59]	0.77	32.91	0.20
	HSCNN+ [26]	4.65	105.84	0.26
	HRNet [28]	31.70	41.95	0.24
	SIGNet [60]	0.84	-	0.11
	MPRNet [61]	3.62	38.67	0.17
	HINet [62]	5.22	12.29	0.15
	AGD-Net [63]	0.77	49.01	0.12
GAN-based	ADV-GAN [30]	11.47	14.17	0.05
Transformer-based	Mst++ [29]	1.62	41.49	0.23
Diffusion-based	SCDM [54]	77.51	33.93	123.77
	DPDM_stage1	1.60	0.00	53.20
	DPDM_stage2	78.28	0.00	58.33

results in substantially slower inference speeds compared to CNN, GAN, and Transformer-based counterparts. Notably, as a single-sample optimization model, the DPDM requires no external training data and thus incurs negligible training time. When compared to the SCDM, another diffusion-based method, the DPDM achieves superior training and inference speeds, albeit at the cost of a greater parameter volume. This result highlights a clear design tradeoff, prioritizing computational efficiency over model compactness.

D. Ablation Studies

In this section, we further investigate the role of the main components in the DPDM. First, we categorize the components of the DPDM into a single-sample optimization denoising net (SDN), PIN, a convolutional block (CB), and a spectral attention block (SAB). The SDN and PIN are incorporated in the first stage, whereas the CB and SAB constitute the SRM in the second stage. Subsequently, we conducted additional experiments to validate the parameter settings and the number of stacked SRMs in the second stage.

1) Main Design of the DPDM: We primarily use the Washington DC Mall dataset to conduct ablation experiments on key modules. Among the components, the SDN is the component of the first-stage diffusion model, primarily used for single-sample noise prediction. PIN is a core component that transforms the input MSI into a structured starting point for the diffusion process. It achieves this by leveraging a spectral basis derived from the SRF and subsequently guides each step of the PDM's reverse process. The SRM utilizes a multilayer network architecture and adopts a single-sample semi-supervised optimization strategy to enhance the spectral fidelity of the reconstructed HSIs. It comprises parallel convolutional and attention blocks, whose respective contributions are validated through ablation experiments.

The experimental results, presented in Table V, demonstrate that the SDN enhances the model's performance across multiple evaluation metrics. The integration of PIN further improves overall performance by leveraging its embedded prior information to guide and stabilize the model's reverse process. It is noteworthy that in the reverse diffusion process, the features from PIN actively guide and influence the evolution of the process variable.

In the second stage, both the CB and SAB contribute to improvements in various performance metrics, indicating the effectiveness of the proposed modular design. Overall, the ablation studies confirm that each core component of the model plays a critical role in the reconstruction of HSIs. The proposed dual-stage architecture significantly enhances the diffusion model's performance in SSR tasks.

2) Parameter Efficiency of the DPDM: In this part, we investigate the impact of varying diffusion steps, latent channel dimensions, and the number of stacked SRMs on the overall model performance. Other experimental settings are kept constant to ensure a fair comparison of the examined variables.

The diffusion step configuration constitutes a critical parameter governing the generation performance of diffusion models. To systematically investigate the influence of step counts on the PDM, we conducted an ablation study on the Washington DC Mall dataset with three distinct configurations: 50, 250, and 500 steps. As demonstrated in Table VI, the 500-step configuration achieves optimal performance across most evaluation metrics, including the MRAE, RMSE, and PSNR. This empirical evidence indicates that sufficient sampling steps are required to ensure reconstruction quality. The observed performance peak at 500 steps indicates an effective tradeoff between computational time and reconstruction accuracy for our implementation.

Moreover, we explore the impact of different latent channel dimensions of the SRM, setting the dimensions to 4, 8, 16, and 32, respectively. We select latent channels as the variable because they directly influence the model's representational capacity and the number of parameters. The experimental results are shown in Table VII. Interestingly, increasing the latent channels did not lead to better hyperspectral reconstruction. However, when the number of latent dimension channels reaches 8, the model yields suboptimal performance, and further increasing the channel number may even degrade performance across the evaluation metrics. Considering the tradeoff between performance gains and model complexity, we select 8 as the final configuration. In Fig. 12, we present a clearer depiction of the relationship between the model's parameters and generated images as the latent channel dimension changes.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:23:09 UTC from IEEE Xplore. Restrictions apply.

TABLE V
ABLATION STUDY OF THE MODULE COMBINATION OF THE DPDM ON THE WASHINGTON DC MALL DATASET

Backbone	Stage1	Stage2			MRAE ↓	RMSE ↓	SAM ↓	PSNR ↑
	SDN	PIN	CB	SAB				
Baseline					0.0963	0.0553	0.0515	23.8548
DPDM	✓				0.0927	0.0532	0.0544	23.9689
DPDM	✓	✓			0.0594	0.0341	0.0621	26.8197
DPDM	✓	✓	✓		0.0446	0.0254	0.0404	27.5768
DPDM	✓	✓	✓	✓	0.0402	0.0229	0.0387	29.6054

TABLE VI
ABLATION STUDY ON THE NUMBER OF DIFFUSION STEPS IN THE DPM USING THE WASHINGTON DC MALL DATASET. ‡ DENOTES THE SELECTED CONFIGURATION

Steps	MRAE ↓	RMSE ↓	SAM ↓	PSNR ↑
50	0.0804	0.0457	0.0839	24.9326
250	0.0604	0.0347	0.0706	26.6274
500‡	0.0594	0.0341	0.0621	26.8197

TABLE VII
ABLATION STUDY OF THE LATENT CHANNEL DIMENSION OF THE SRM ON THE WASHINGTON DC MALL DATASET. ‡ DENOTES THE SELECTED CONFIGURATION

Latent Channel	MRAE ↓	RMSE ↓	SAM ↓	PSNR ↑
4	0.0408	0.0233	0.0392	29.4681
8‡	0.0402	0.0229	0.0387	29.6054
16	0.0433	0.0247	0.041	28.7667
32	0.0394	0.0225	0.038	29.7149

TABLE VIII
ABLATION STUDY OF THE NUMBER OF STACKED SRMS ON THE WASHINGTON DC MALL DATASET. ‡ DENOTES THE SELECTED CONFIGURATION

Number of stacked SRM	MRAE ↓	RMSE ↓	SAM ↓	PSNR ↑
1	0.0656	0.0372	0.0647	26.8831
2‡	0.0402	0.0229	0.0387	29.6054
3	0.0429	0.0245	0.0418	28.8286
4	0.0466	0.0266	0.0437	27.8139

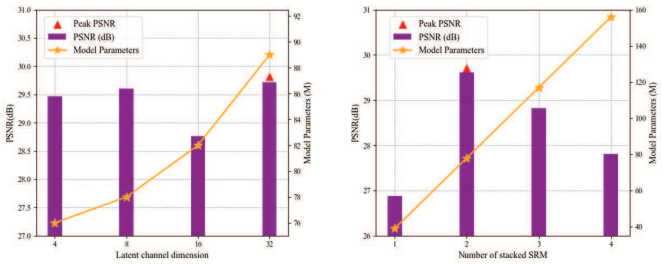


Fig. 12. (Left) Ablation of different latent channels. (Right) Ablation of the number of stacked SRMs.

Furthermore, we investigate the impact of varying the number of stacked SRMs on HSI generation, with the number of stacks set to 1, 2, 3, and 4. The experimental results are presented in Table VIII; the model's performance does not monotonically improve with an increasing number of stacks. In contrast, the best performance is achieved when the number of stacks is set to 2. This indicates that the additional parameters introduced by excessively stacking modules may negatively affect the model's learning capability. A more detailed comparison of model performance and parameter complexity with respect to different stack numbers is illustrated in Fig. 12.

V. CONCLUSION

In this article, we introduced the DPDM, a novel framework composed of two cascaded modules: a PDM and an SRM. The PDM pioneers the use of PIN, which replaces the standard random noise input. By providing PIN as both an initial state and a continuous guide for the reverse diffusion process, the PDM significantly enhances spectral feature learning. The module's performance is further elevated by a redesigned noise prediction network tailored for superior spectral representation. Complementing the PDM, the SRM specifically mitigates noise and artifacts introduced during the diffusion process. It refines the coarse output by restoring fine-grained spatial–spectral details, thereby boosting the overall fidelity of the generated HSI. Comprehensive comparisons on three benchmark datasets demonstrate the DPDM's strong performance. However, we acknowledge a primary limitation of our model. Like many diffusion-based methods, the DPDM has high computational costs and long sampling times. This constrains its utility for real-time applications. Therefore, the DPDM is particularly well-suited for offline tasks where high precision is paramount, even with limited data. Such applications include disaster impact assessment and agricultural monitoring. Our future work will focus on enhancing the model's sampling efficiency to broaden its applicability.

REFERENCES

[1] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.

[2] L. C. Parra, C. D. Spence, P. Sajda, A. Ziehe, and K. Müller, "Unmixing hyperspectral data," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 12, 1999, pp. 942–948.

[3] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2019.

[4] A. A. Nielsen, "The regularized iteratively reweighted MAD method for change detection in multi- and hyperspectral data," *IEEE Trans. Image Process.*, vol. 16, no. 2, pp. 463–478, Feb. 2007.

[5] Y. Wang, Q. Yuan, T. Li, H. Shen, L. Zheng, and L. Zhang, "Large-scale MODIS AOD products recovery: Spatial–temporal hybrid fusion considering aerosol variation mitigation," *ISPRS J. Photogramm. Remote Sens.*, vol. 157, pp. 1–12, Nov. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:23:09 UTC from IEEE Xplore. Restrictions apply.

[6] Q. Wei, J. Bioucas-Dias, N. Dobigeon, and J.-Y. Tourneret, "Hyperspectral and multispectral image fusion based on a sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3658–3668, Jul. 2015.

[7] X. Sun, J. Gao, and Y. Yuan, "Alignment and fusion using distinct sensor data for multimodal aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5626811.

[8] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Spectral–spatial hyperspectral image segmentation using subspace multinomial logistic regression and Markov random fields," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 3, pp. 809–823, Mar. 2012.

[9] Q. Wang, Y. Jia, W. Huang, J. Gao, and Q. Li, "Embedding generalized semantic knowledge into few-shot remote sensing segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5603413.

[10] Y. Liu, G. Gao, and Y. Gu, "Tensor matched subspace detector for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 4, pp. 1967–1974, Apr. 2017.

[11] J. Gao, D. Zhang, F. Wang, L. Ning, Z. Zhao, and X. Li, "Combining SAM with limited data for change detection in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5614311.

[12] J. He et al., "Spectral super-resolution meets deep learning: Achievements and challenges," *Inf. Fusion*, vol. 97, Sep. 2023, Art. no. 101812.

[13] H. Yu, Z. Ling, K. Zheng, L. Gao, J. Li, and J. Chanussot, "Unsupervised hyperspectral and multispectral image fusion with deep spectral–spatial collaborative constraint," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534114.

[14] L. Gao, J. Li, K. Zheng, and X. Jia, "Enhanced autoencoders with attention-embedded degradation learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Apr. 2023, Art. no. 5509417.

[15] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218.

[16] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.

[17] J. Li, K. Zheng, Z. Li, L. Gao, and X. Jia, "X-shaped interactive autoencoders with cross-modality mutual learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518317.

[18] R. Dian, S. Li, L. Fang, T. Lu, and J. M. Bioucas-Dias, "Nonlocal sparse tensor factorization for semiblind hyperspectral and multispectral image fusion," *IEEE Trans. Cybern.*, vol. 50, no. 10, pp. 4469–4480, Oct. 2020.

[19] B. Arad and O. Ben-Shahar, "Sparse recovery of hyperspectral signal from natural RGB images," in *Proc. Eur. Conf. Comput. Vis.*Cham, Switzerland: Springer, 2016, pp. 19–34.

[20] J. Wu, J. Aeschbacher, and R. Timofte, "In defense of shallow learned spectral reconstruction from RGB images," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2017, pp. 471–479.

[21] K. Fotiadou, G. Tsagkatakis, and P. Tsakalides, "Spectral super resolution of hyperspectral images via coupled dictionary learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2777–2797, May 2019.

[22] H.-F. Yan, Y.-Q. Zhao, J. C.-W. Chan, and S. G. Kong, "Spectral super-resolution based on dictionary optimization learning via spectral library," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5400216.

[23] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[24] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[25] I. Goodfellow et al., "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.

[26] Z. Shi, C. Chen, Z. Xiong, D. Liu, and F. Wu, "HSCNN+: Advanced CNN-based hyperspectral recovery from RGB images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 1052–10528.

[27] Z. Xiong, Z. Shi, H. Li, L. Wang, D. Liu, and F. Wu, "HSCNN: CNN-based hyperspectral image recovery from spectrally undersampled projections," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2017, pp. 518–525.

[28] Y. Zhao, L.-M. Po, Q. Yan, W. Liu, and T. Lin, "Hierarchical regression network for spectral reconstruction from RGB images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 1695–1704.

[29] Y. Cai et al., "MST++: Multi-stage spectral-wise transformer for efficient spectral reconstruction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 745–755.

[30] A. Alvarez-Gila, J. Van De Weijer, and E. Garrote, "Adversarial networks for spatial context-aware spectral image reconstruction from RGB," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2017, pp. 480–490.

[31] P. Dhariwal and A. Nichol, "Diffusion models beat GANs on image synthesis," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 8780–8794.

[32] C. Saharia et al., "Photorealistic text-to-image diffusion models with deep language understanding," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 36479–36494.

[33] C. Saharia, J. Ho, W. Chan, T. Salimans, D. J. Fleet, and M. Norouzi, "Image super-resolution via iterative refinement," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4713–4726, Apr. 2023.

[34] H. Zhang, W. He, L. Zhang, H. Shen, and Q. Yuan, "Hyperspectral image restoration using low-rank matrix recovery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4729–4743, Aug. 2014.

[35] M. Aharon, M. Elad, and A. Bruckstein, "K-SVD: An algorithm for designing overcomplete dictionaries for sparse representation," *IEEE Trans. Signal Process.*, vol. 54, no. 11, pp. 4311–4322, Nov. 2006.

[36] X. X. Zhu, C. Grohnfeldt, and R. Bamler, "Exploiting joint sparsity for pansharpening: The J-SparseFI algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 2664–2681, May 2016.

[37] B. Zhao, H. Li, X. Lu, and X. Li, "Reconstructive sequence-graph network for video summarization," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 5, pp. 2793–2801, May 2022.

[38] J. Han, D. Zhang, S. Wen, L. Guo, T. Liu, and X. Li, "Two-stage learning to predict human eye fixations via SDAEs," *IEEE Trans. Cybern.*, vol. 46, no. 2, pp. 487–498, Feb. 2016.

[39] H. Guo, J. Gao, and Y. Yuan, "Balanced density regression network for remote sensing object counting," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5625013.

[40] X. Han, H. Zhang, J.-H. Xue, and W. Sun, "A spectral–spatial jointed spectral super-resolution and its application to HJ-1A satellite images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[41] Y. B. Can and R. Timofte, "An efficient CNN for spectral reconstruction from RGB images," 2018, *arXiv:1804.04647*.

[42] S. Koundinya et al., "2D–3D CNN based architectures for spectral reconstruction from RGB images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 957–9577.

[43] L. Yan, X. Wang, M. Zhao, M. Kaloorazi, J. Chen, and S. Rahardja, "Reconstruction of hyperspectral data from RGB images with prior category information," *IEEE Trans. Comput. Imag.*, vol. 6, pp. 1070–1081, 2020.

[44] H. Cao et al., "A survey on generative diffusion model," 2022, *arXiv:2209.02646*.

[45] K. Jiang, Z. Wang, P. Yi, G. Wang, T. Lu, and J. Jiang, "Edge-enhanced GAN for remote sensing image superresolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5799–5812, Aug. 2019.

[46] R. Wang, B. Jiang, C. Yang, Q. Li, and B. Zhang, "MAGAN: Unsupervised low-light image enhancement guided by mixed-attention," *Big Data Mining Anal.*, vol. 5, no. 2, pp. 110–119, Jun. 2022.

[47] J. Ho, A. N. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 6840–6851.

[48] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," 2020, *arXiv:2010.02502*.

[49] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 10684–10695.

[50] A. Hertz, R. Mokady, J. Tenenbaum, K. Aberman, Y. Pritch, and D. Cohen-Or, "Prompt-to-prompt image editing with cross attention control," 2022, *arXiv:2208.01626*.

[51] C. Wu, D. Wang, Y. Bai, H. Mao, Y. Li, and Q. Shen, "HSR-Diff: Hyperspectral image super-resolution via conditional diffusion models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 7060–7070.

[52] Y. Xiao, Q. Yuan, K. Jiang, J. He, X. Jin, and L. Zhang, "EDiffSR: An efficient diffusion probabilistic model for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5601514.

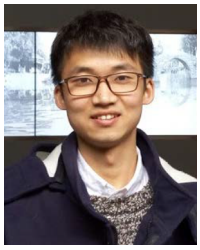
Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:23:09 UTC from IEEE Xplore. Restrictions apply.

[53] L. Liu, B. Chen, H. Chen, Z. Zou, and Z. Shi, "Diverse hyperspectral remote sensing image synthesis with diffusion models," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5532616.
[54] B. Chen, L. Liu, C. Liu, Z. Zou, and Z. Shi, "Spectral-cascaded diffusion model for remote sensing image spectral super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5528414.
[55] V. Lempitsky, A. Vedaldi, and D. Ulyanov, "Deep image prior," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 9446–9454.
[56] B. Arad et al., "NTIRE 2020 challenge on spectral reconstruction from an RGB image," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 1806–1822.
[57] B. Lim, S. Son, H. Kim, S. Nah, and K. M. Lee, "Enhanced deep residual networks for single image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 1132–1140.
[58] J. Li, C. Wu, R. Song, Y. Li, and F. Liu, "Adaptive weighted attention network with camera spectral sensitivity prior for spectral reconstruction from RGB images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 1894–1903.
[59] J. He, J. Li, Q. Yuan, H. Shen, and L. Zhang, "Spectral response function-guided deep optimization-driven network for spectral super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 9, pp. 4213–4227, Sep. 2022.
[60] J. Li, S. Du, R. Song, Y. Li, and Q. Du, "Progressive spatial information-guided deep aggregation convolutional network for hyperspectral spectral super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 1, pp. 1677–1691, Jan. 2025.
[61] S. W. Zamir et al., "Multi-stage progressive image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14821–14831.
[62] L. Chen, X. Lu, J. Zhang, X. Chu, and C. Chen, "HINet: Half instance normalization network for image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 182–192.
[63] Z. Zhu, H. Liu, J. Hou, S. Jia, and Q. Zhang, "Deep amended gradient descent for efficient spectral reconstruction from single RGB images," *IEEE Trans. Comput. Imag.*, vol. 7, pp. 1176–1188, 2021.



Zengyi Li received the M.S. degree in computer technology from the National Space Science Center, Chinese Academy of Sciences, Beijing, China, in 2024. He is currently pursuing the Ph.D. degree with the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an, China.

His research interests include spectral super-resolution and multimodal learning.



Junyu Gao (Member, IEEE) received the B.E. and Ph.D. degrees in computer science and technology from Northwestern Polytechnical University, Xi'an, China, in 2015 and 2021, respectively.

He is currently an Associate Professor with the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University. His research interests include computer vision and pattern recognition.



Yuan Yuan (Senior Member, IEEE) is currently a Full Professor with the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an, China. She has authored or co-authored over 150 papers, including about 100 in reputable journals, such as IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, and conference papers in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), British Machine Vision Conference (BMVC), International Conference on Image Processing (ICIP), and International Conference on Acoustics, Speech and Signal Processing (ICASSP). Her research interests include visual information processing and image/video content analysis.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:23:09 UTC from IEEE Xplore. Restrictions apply.